

The Impact of Anime-Styled Personas on Engagement and Enjoyment in Conversational Educational Agents

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Abstract— Conversational agents are increasingly used in educational contexts to enhance learning through interactive experiences. Focusing on the domain of Human-Computer Interaction (HCI) and Human-Agent Interaction (HAI), this study explores how the use of a stylised, anime-inspired persona in a robot tutor affects user engagement, perceived enjoyment, and learning outcomes among university students. Two tutor personas were designed and deployed on the Furhat robot platform - a traditional, formal agent (Dr.AI) and a playful, anime-themed agent (AI-Chan). Through a high-fidelity prototype and a pilot user study with ten participants, the study examined how differences in verbal tone, visual design, and affective expression influenced learners' interaction experiences. The agents' effectiveness was measured using user engagement and enjoyment surveys, specifically the User Engagement Scale (UES) and custom questions adapted from the IMI enjoyment/interest subscale. The results reveal that users found the anime-style persona more enjoyable and engaging. Despite technical limitations, these findings illustrate promising evidence for the value of persona-driven design in conversational learning agents.

Keywords— *human-robot interaction, engagement, learning outcomes, educational technology, social robotics, embodied conversational agents*

I. INTRODUCTION

The integration of conversational agents into educational environments has introduced new opportunities to improve user engagement and motivation. Traditional educational agents are often designed to be professional and embody a formal, neutral tone. However, research in Human-Computer Interaction (HCI) and Human-Agent Interaction (HAI) suggests that stylistic features in agent personas, such as voice, appearance, and affective behavior, can significantly impact user experience. This paper investigates whether an anime-style persona implementation in a tutor robot can enhance engagement and perceived enjoyment of students in a university learning context. The study focuses on anime-inspired persona characterization as anime characters are culturally familiar to many young adults, and are often associated with emotionally resonant, expressive behavior – a trait attributed to a more engaging interaction. The study's hypothesis assumes that an anime-inspired persona may make the learning experience more immersive and enjoyable,

resulting in higher user engagement and learned content retention in the setting of a self-directed study with a conversational robot tutor. The null hypothesis assumes that the use of an anime persona in a robot tutor will not have an impact on user engagement and learned content retention. To test the hypotheses, the experiment features two distinct robot tutor personas of a robot tutor provided by the Furhat humanoid social robot platform. One of the robot tutors is AI-Chan – an anime-style tutor with expressive voice, tone and playful language, and Dr.AI, a traditional, formal, neutral-tone tutor. Both personas deliver the same educational content about the history of artificial intelligence. A pilot study is conducted with ten university students, and their perception of the learning experience with the two different robot tutors is evaluated. The evaluation methods include a user feedback survey consisting of the User Engagement Scale (UES) and custom questions adapted from the IMI enjoyment/interest subscale.

Recent studies in educational technology and HCI provide evidence that persona design has a positive impact on user engagement and motivation in learning tasks. Drobnyak et al. [5] highlight that interactive, personalized educational agents, including those with stylized personas, can significantly improve learning outcomes. Mahler and Mayer [1] found that anime-based content led to higher user motivation, enjoyment, and retention compared to traditional methods of agent interaction, while Tainaka et al. [3] and Al-Sada et al. [2] showed anime's effectiveness in improving conversational agents' interactions with users through increased emotional expressiveness. Smith and Roberts [6] support the use of generative AI to reduce academic boredom, reinforcing this study's hypothesis that anime-style conversational agents can make learning more enjoyable and motivating for university students. Despite this growing evidence, few studies directly compare anime personas with standard formal agents in robot tutoring environments – an important research gap that this study aims to address.

II. METHODS

The study uses a within-subjects design, with every participant taking part in both conditions. There are two distinct conditions: a conversation with the anime persona, known as AI-Chan, which has an anime-style face and speaks with the idiolect of an anime character, and the formal condition, with the robot, known as Dr AI, being formal and displaying behaviour more akin to a standard classroom teacher. These serve as the experimental and control conditions, respectively. Each of these teaches something related to the history of AI. Hence, the independent variable (IV) is the type of robot persona being used. Potential uncontrolled variables include individual differences in the participant's familiarity with and appreciation for anime culture, as well as a variability in the level of comfort each participant experiences when interacting with a robot. To mitigate these effects, background information is collected on each participant to balance results. Moreover, each participant is part of the artificial intelligence bachelor and therefore should all have a relatively high familiarity with robots, which balances further each participant's comfort with them. The apparatus consists of the Furhat robot, a laptop, digital consent forms, and digital Google Form surveys. Each participant was first provided with a consent form to sign, before sitting in front of a laptop with the Furhat software activated. The participant then interacts with each robot, asking questions in relation to AI. After completing the interaction, they fill in a Google Form which asks them a variety of questions with regards to their enjoyment in the interaction, thoughts on each robot, as well as feedback for improving the software in the future. They are also asked if they enjoy anime, as this may bias their enjoyment when interacting with the anime persona. The target sample comprises of university students in AI aged from 18-25 and were collected through convenience sampling. A minimum of 8 participants are required. All the data to be collected is via the Google Form that each participants fill in. They must rate their experiences from 1-7 in terms of enjoyment for each robot, as well as a separate rating for clarity and understanding of explanations provided by each robot which serve as the two variables being measured. Overall, both qualitative and quantitative data is collected, with the ratings being quantitative and the feedback being qualitative. This data, particularly quantitative data, will allow the hypothesis that engagement is increased due to the anime robot, to be validated by comparing the ratings on the formal robot versus the anime robot. Finally, the data is stored securely in a google spreadsheet protected by a password and only accessible by the researchers.

III. RESULTS

A. Figures

Do you like / watch anime?

12 responses

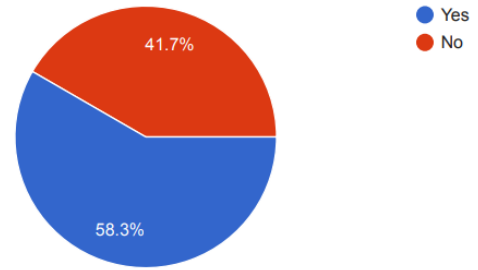


Fig. 1 Responses by participants in regard to enjoyment of anime

How much did you like interacting with Dr. AI?

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12 responses

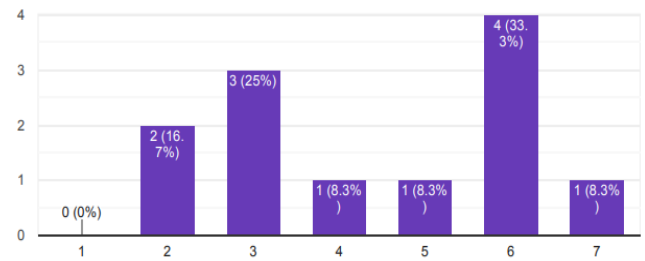


Fig. 2: Bar chart for responses for engagement with formal robot, rated 1-7

How much did you like interacting with Dr. AI?

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12 responses

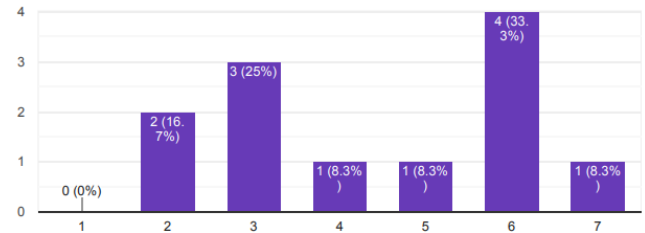


Fig. 3: Bar chart for responses for engagement with anime robot,

i

How clear and useful were Dr. AI's explanations?

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12 responses

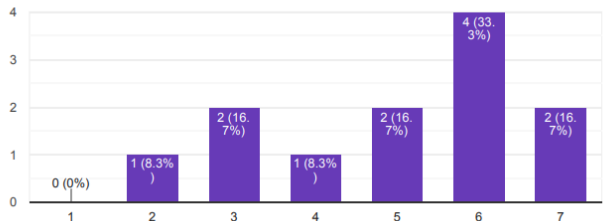


Fig. 4 :Bar chart for responses for clarity of formal robot

How clear and useful were AI-chan's explanations?

12 responses

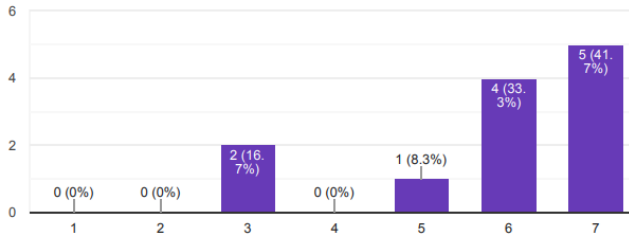


Fig. 5: Bar chart for responses for clarity of anime robot

Who would you prefer to learn from in the future?

12 responses

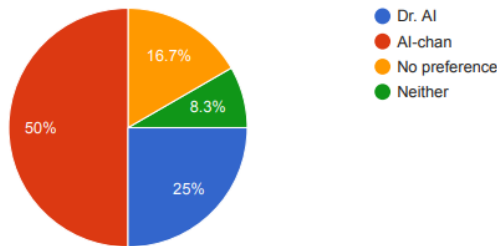


Fig. 6: Pie chart showing preferred robot for explanations and enjoyment

B. Statistical analysis

Given that the data is from a within-subject design and does not necessarily follow a normal distribution, it is appropriate to use a Wilcoxon signed-rank test for statistical analysis. This test will be done on the ratings for clarity of responses of both robots and the ratings for engagement with each robot. Performing the test, results found that for the engagement enjoyment of the formal robot versus the anime robot. The Wilcoxon signed-rank test for engagement of the formal robot versus the anime robot yielded a statistic of 8.5 with a p-value of 0.095.

For clarity and understanding ratings, the Wilcoxon statistic was 12.5 with a p-value of 0.44. Statistical analysis therefore indicates no significant findings, given that the p-value is above 0.05 in both variables measured. While these results did not reach statistical significance ($p > 0.05$), the descriptive data and participant preferences suggest a trend toward increased engagement and enjoyment with the anime-styled agent.

• Discussion

While statistical analysis on the quantitative data did not show a significant difference in engagement and clarity between the two robots, other data does show that participants did appear to prefer the anime robot overall. For example, the pie chart shows a 50% preference for the anime robot against a 25% preference for the formal robot. Moreover, qualitative data shows a distinct preference for AI-Chan, with comments such as 'It was much more engaging than Dr. AI' and 'It was

engaging and exciting to talk to an anime teacher'. All in all, despite a lack of statistical significance being analyzed, it can still be found that these findings ultimately support an anime style education.

The findings of this research add to the existing research that demonstrates the importance of agent design in user engagement in learning environments. This is like previous research that shows the emotional impact of an anime-style agent in human-robot interaction [2], [3]. Animated facial expressions and speech in AI-Chan are in line with research that illustrates that emotional richness improves user-agent interaction [3], [4].

Whereas past studies have investigated the effect of anime-style embodiments [2], our study operationalizes these stylistic elements in a learning context with the assistance of advanced interactions. In comparison to the work to [4], in which emotional expressiveness in chatbots played a central role, our results contribute to an explanation of how these design elements additionally influence content understanding and the learning process when they are used by embodied agents in live educational interactions.

Additionally, whereas [5] systematically reviewed the application of personas in AI learning, our research adds empirical support to the literature. In contrast to longitudinal studies like [6], which focus on ethics in AI and long-term retention, our research focuses on short-term levels of engagement within a single session.

Overall, these findings indicate that the use of culturally relevant design features, such as personas based on anime, can have a tremendous impact on user experience in learning contexts. Future studies should examine how long such effects last and how generalizable they are to populations with different demographics and traits.

IV. CONCLUSION

This study examined how different robot tutor personas can affect user engagement and enjoyment and influence their learning outcomes. A formal instructional agent (Dr. AI) was compared with an anime-styled agent (AI-Chan) using a high-fidelity prototype. To evaluate the user experience, the participants were asked to complete surveys after interacting with both of the agents. These surveys included the User Engagement Scale, custom questions, and the IMI. It was revealed that users responded more positively to the anime agent, perceiving it as more engaging and comprehensible. These findings, while not statistically significant, support the hypothesis that using an anime-style persona in a tutor robot will enhance engagement and perceived enjoyment of university students. These findings contribute to the ongoing research in Human-Computer interaction by proving that character design impacts user experience. Future research should involve examining the long-term effect of persona design on the users' motivation and retention across longer research durations because this study concentrated primarily on short-term engagement and enjoyment. The sample

population could cover varied age groups and cultural backgrounds. With advancing education technologies, the integration of culturally responsive and emotionally dense personas into artificially intelligent agents may be crucial in fostering denser and more engaging learning experiences.

STATEMENT OF CONTRIBUTION

Karina was responsible for writing the abstract and introduction and contributed by gathering part of the respondents.

Suhani contributed by writing the discussion and conclusion sections, as well as formatting and compiling references and citations.

Iain worked on the methodology and results sections.

Amina was responsible for the technical components of the project and also contributed to data collection by gathering part of the respondents.

REFERENCES

- [1] L. S. Mahler and R. E. Mayer, "Anime as a medium for science learning," *J. Comput. Assist. Learn.*, vol. 40, no. 2, pp. 787–796, 2024.
<https://doi.org/10.1111/jcal.12908>

- [2] M. Al-Sada, P. C. Yang, C. C. Chiu, T. P. Tomo, M. Y. Saraiji, T. Ogata, and T. Nakajima, "From Anime to Reality: Embodying an Anime Character as a Humanoid Robot," in *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*, May 2021, pp. 1–5.

<https://doi.org/10.1145/3411763.3451543>

- [3] K. Tainaka, T. Horiuchi, H. F. Putra, and T. Kobayashi, "Enriching Emotional Expressiveness in Embodied Conversational Agents Using Anime Effects," *IEEE Access*, 2025.

[10.1109/ACCESS.2025.3559082](https://doi.org/10.1109/ACCESS.2025.3559082)

- [4] Y. T. Wan, C. C. Chiu, K. W. Liang, and P. C. Chang, "Midoriko chatbot: LSTM-based emotional 3D avatar," in *2019 IEEE 8th Global Conference on Consumer Electronics (GCCE)*, Oct. 2019, pp. 937–940.

[10.1109/GCCE46687.2019.9015303](https://doi.org/10.1109/GCCE46687.2019.9015303)

- [5] A. Drobnyak, I. Boticki, S. E. O. W. Peter, and K. A. H. N. Ken, "Learning with conversational AI and personas: A systematic literature review," in *Proc. Int. Conf. Comput. Educ.*, Dec. 2023.

<https://doi.org/10.58459/icce.2023.1390>

- [6] C. Smith and E. Roberts, "Unveiling Academic Boredom: Leveraging Generative AI for Ethical and Engaging Learning Experiences in Higher Education," 2024.

<https://salford-repository.worktribe.com/output/2951474>